

Reflection:

This was an interesting experiment and I definitely learned a lot from the process. Implementing any kind of model manually provides two primary benefits, full control of the process and better understanding of the results. This is the first neural network I have ever programmed, so the learning experience alone was worth the effort. I can easily see how this could occupy multiple semester long classes.

MSE is not the natural loss for a probability-producing sigmoid output -- its gradient shrinks when the prediction is far from the true label but the sigmoid is saturated (near 0 or 1), which can slow learning when the model is the most wrong. Binary cross-entropy is generally preferred for classification because its gradient does not have this issue and more directly penalizes confident wrong predictions. MSE also does not produce a calibrated probability the way a cross-entropy-trained model's output would, so the "probability-like" scores here should be read as relative rankings more than true probabilities. It does have the advantage of being computationally light and thus relatively fast.

Of the things we know, the network doesn't consider the university, program, time of year of the application, or anything in the comments. That said, there are a lot of unknowns in the dataset that could affect the acceptance rates. Some of these things are tangible numbers that could play a part, like how many seats are available at a university vs the number of acceptances as well as number we simply are not given by the individuals filling out the entry (no GPA, GRE, GRE V, or GRE AW data for example). Adding more of these metrics could help strengthen the model overall. But there still remains aspects that the model will never really be able to touch.

A lot of the missing pieces are intangibles that would be difficult to quantify mathematically, such as essay submissions, interviews, or relevant work/life experience. These intangibles are the wild cards that make the dataset only so accurate. From the data I pulled the test was producing what reflected as about 70% accuracy. That's based on a binary split on a value I would interpret more as saying, "you rank pretty high/low compared to other people who have submitted info", as opposed to "this is 70% accurate". The reflection of accuracy is not an entirely true idea based on the methodology.

"Accuracy" could be increased by gaining enough data to focus on specific universities and specific programs. Using data reported directly from the university would be preferred. This would give a much more accurate view of the specifics required to gain entry into a program. It could even include concrete numbers for some of the aforementioned intangibles (depending how the university uses some of these items, perhaps by assigning a score to an essay or interview). Focusing on these two aspects would also eliminate individuals that apply to multiple programs that could also be skewing the data. For example, an individual with a certain GRE applying to five separate programs currently shows up as five separate individuals. When they only receive one acceptance, it skews the GRE weighting toward Rejected.

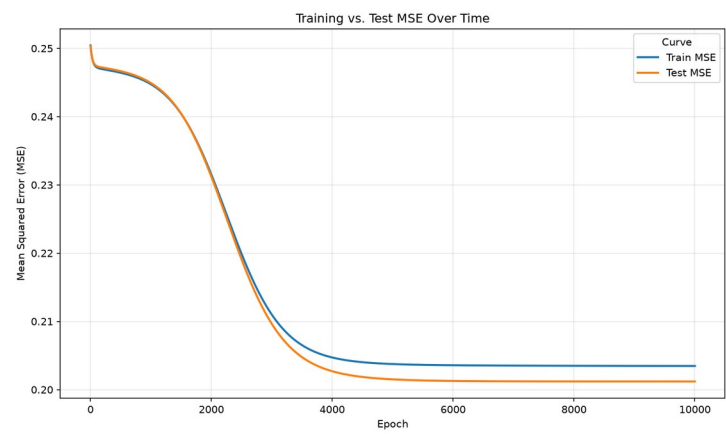
This is a fun experiment, but the premise has a lot of blindspots that can not really be filled.

Artificial Applicant Findings:

Applicant	GPA	GRE	GRE-V	GRE-AW	Degree	Citizenship	Pred Prob	Pred Status
Strong	3.90	168	165	5.0	PhD	International	0.3038	Rejected
Weaker	3.30	150	148	3.5	PhD	US	0.2869	Rejected
Strong	3.85	165	162	4.5	Masters	International	0.7441	Accepted
Weaker	3.20	148	145	3.0	Masters	US	0.6946	Accepted

Both PhD applicants (strong and weaker profile) are predicted Rejected (~0.30 and ~0.29 probability), while both Masters applicants are predicted Accepted (~0.74 and ~0.69 probability). Within each degree type the strong-profile applicant does get a somewhat higher predicted probability than the weaker-profile one, so GPA/GRE are having some effect -- but that effect is small compared to the gap between PhD and Masters applicants. In other words, ``ms_vs_phd`` appears to be the dominant driver of these four predictions, with the numeric features acting more as a secondary adjustment than a primary driver. Whether that reflects a genuine pattern in the real GradCafe data (PhD programs being more competitive/selective than Masters programs, for instance) or the network leaning on the single strongest, cleanest, always-present binary feature instead of the noisier, heavily-missing numeric ones, is an interesting question that has no immediate answer.

mse_curve:



Final Evaluation Results:

Rows used after filtering: 38889

Train size: 31111

Test Size: 7778

Best Epoch: 10000

Best test MSE: 0.201197

Final train accuracy: 0.7138

Final test accuracy: 0.7184

Final training accuracy (0.7138) and final test accuracy (0.7184) are close to one another and training/test MSE tracked closely throughout rather than diverging. This shows no sign of overfitting.

The majority-class baseline (always predicting "Rejected") is about 55.1% on the test set. A final test accuracy of ~71.8% is roughly 16.7 percentage points above that baseline -- a meaningful, not marginal, improvement, and one that appears stable rather than unstable, since MSE and accuracy moved smoothly across epochs with no large swings.

Six features (GPA, three GRE scores, degree type, and citizenship) leave out most of what actually drives a real admissions decision -- program or department, research fit, letters of recommendation, statement of purpose, prior publications, and program-specific competitiveness, none of which are present here. A large share of GPA/GRE values were also missing (and had to be median-filled), or worse, contained values that were completely out of bounds. So while this feature set clearly carries genuine signal (clearing the baseline by 16+ points), it is not sufficient on its own for a realistic, high-confidence admissions predictor.

Training Output:

--- Training ---

Epoch 100/10000	- Train MSE: 0.247231, Test MSE: 0.247419, Test Acc: 0.5509
Epoch 200/10000	- Train MSE: 0.246947, Test MSE: 0.247182, Test Acc: 0.5509
Epoch 300/10000	- Train MSE: 0.246786, Test MSE: 0.247027, Test Acc: 0.5509
Epoch 400/10000	- Train MSE: 0.246616, Test MSE: 0.246856, Test Acc: 0.5509
Epoch 500/10000	- Train MSE: 0.246423, Test MSE: 0.246659, Test Acc: 0.5509
Epoch 600/10000	- Train MSE: 0.246199, Test MSE: 0.246428, Test Acc: 0.5509
Epoch 700/10000	- Train MSE: 0.245931, Test MSE: 0.246153, Test Acc: 0.5509
Epoch 800/10000	- Train MSE: 0.245608, Test MSE: 0.245820, Test Acc: 0.5509
Epoch 900/10000	- Train MSE: 0.245216, Test MSE: 0.245415, Test Acc: 0.5509
Epoch 1000/10000	- Train MSE: 0.244739, Test MSE: 0.244922, Test Acc: 0.5509
Epoch 1100/10000	- Train MSE: 0.244159, Test MSE: 0.244322, Test Acc: 0.5509
Epoch 1200/10000	- Train MSE: 0.243457, Test MSE: 0.243595, Test Acc: 0.5509
Epoch 1300/10000	- Train MSE: 0.242611, Test MSE: 0.242719, Test Acc: 0.5509
Epoch 1400/10000	- Train MSE: 0.241601, Test MSE: 0.241674, Test Acc: 0.5509
Epoch 1500/10000	- Train MSE: 0.240410, Test MSE: 0.240439, Test Acc: 0.5509
Epoch 1600/10000	- Train MSE: 0.239021, Test MSE: 0.238999, Test Acc: 0.5508
Epoch 1700/10000	- Train MSE: 0.237429, Test MSE: 0.237348, Test Acc: 0.5512
Epoch 1800/10000	- Train MSE: 0.235634, Test MSE: 0.235485, Test Acc: 0.5517
Epoch 1900/10000	- Train MSE: 0.233650, Test MSE: 0.233424, Test Acc: 0.5561
Epoch 2000/10000	- Train MSE: 0.231502, Test MSE: 0.231192, Test Acc: 0.6620
Epoch 2100/10000	- Train MSE: 0.229228, Test MSE: 0.228825, Test Acc: 0.7157
Epoch 2200/10000	- Train MSE: 0.226875, Test MSE: 0.226374, Test Acc: 0.7178
Epoch 2300/10000	- Train MSE: 0.224497, Test MSE: 0.223892, Test Acc: 0.7184
Epoch 2400/10000	- Train MSE: 0.222150, Test MSE: 0.221436, Test Acc: 0.7184
Epoch 2500/10000	- Train MSE: 0.219886, Test MSE: 0.219061, Test Acc: 0.7184
Epoch 2600/10000	- Train MSE: 0.217748, Test MSE: 0.216814, Test Acc: 0.7184
Epoch 2700/10000	- Train MSE: 0.215773, Test MSE: 0.214729, Test Acc: 0.7183
Epoch 2800/10000	- Train MSE: 0.213982, Test MSE: 0.212831, Test Acc: 0.7183
Epoch 2900/10000	- Train MSE: 0.212387, Test MSE: 0.211133, Test Acc: 0.7187
Epoch 3000/10000	- Train MSE: 0.210988, Test MSE: 0.209635, Test Acc: 0.7187
Epoch 3100/10000	- Train MSE: 0.209778, Test MSE: 0.208332, Test Acc: 0.7187
Epoch 3200/10000	- Train MSE: 0.208743, Test MSE: 0.207210, Test Acc: 0.7187
Epoch 3300/10000	- Train MSE: 0.207866, Test MSE: 0.206252, Test Acc: 0.7187
Epoch 3400/10000	- Train MSE: 0.207130, Test MSE: 0.205441, Test Acc: 0.7187
Epoch 3500/10000	- Train MSE: 0.206515, Test MSE: 0.204758, Test Acc: 0.7187
Epoch 3600/10000	- Train MSE: 0.206004, Test MSE: 0.204185, Test Acc: 0.7187
Epoch 3700/10000	- Train MSE: 0.205582, Test MSE: 0.203706, Test Acc: 0.7187
Epoch 3800/10000	- Train MSE: 0.205233, Test MSE: 0.203306, Test Acc: 0.7187
Epoch 3900/10000	- Train MSE: 0.204945, Test MSE: 0.202973, Test Acc: 0.7187
Epoch 4000/10000	- Train MSE: 0.204708, Test MSE: 0.202694, Test Acc: 0.7187
Epoch 4100/10000	- Train MSE: 0.204513, Test MSE: 0.202462, Test Acc: 0.7187
Epoch 4200/10000	- Train MSE: 0.204353, Test MSE: 0.202269, Test Acc: 0.7187
Epoch 4300/10000	- Train MSE: 0.204220, Test MSE: 0.202107, Test Acc: 0.7187
Epoch 4400/10000	- Train MSE: 0.204111, Test MSE: 0.201971, Test Acc: 0.7187
Epoch 4500/10000	- Train MSE: 0.204020, Test MSE: 0.201857, Test Acc: 0.7187
Epoch 4600/10000	- Train MSE: 0.203945, Test MSE: 0.201761, Test Acc: 0.7187
Epoch 4700/10000	- Train MSE: 0.203882, Test MSE: 0.201680, Test Acc: 0.7187
Epoch 4800/10000	- Train MSE: 0.203830, Test MSE: 0.201612, Test Acc: 0.7187
Epoch 4900/10000	- Train MSE: 0.203786, Test MSE: 0.201554, Test Acc: 0.7187
Epoch 5000/10000	- Train MSE: 0.203749, Test MSE: 0.201505, Test Acc: 0.7187
Epoch 5100/10000	- Train MSE: 0.203717, Test MSE: 0.201463, Test Acc: 0.7187
Epoch 5200/10000	- Train MSE: 0.203691, Test MSE: 0.201427, Test Acc: 0.7187
Epoch 5300/10000	- Train MSE: 0.203668, Test MSE: 0.201396, Test Acc: 0.7187
Epoch 5400/10000	- Train MSE: 0.203648, Test MSE: 0.201369, Test Acc: 0.7187
Epoch 5500/10000	- Train MSE: 0.203631, Test MSE: 0.201347, Test Acc: 0.7187
Epoch 5600/10000	- Train MSE: 0.203617, Test MSE: 0.201327, Test Acc: 0.7187

Epoch 5700/10000 - Train MSE: 0.203604, Test MSE: 0.201310, Test Acc: 0.7187
Epoch 5800/10000 - Train MSE: 0.203592, Test MSE: 0.201295, Test Acc: 0.7187
Epoch 5900/10000 - Train MSE: 0.203582, Test MSE: 0.201283, Test Acc: 0.7187
Epoch 6000/10000 - Train MSE: 0.203573, Test MSE: 0.201271, Test Acc: 0.7187
Epoch 6100/10000 - Train MSE: 0.203565, Test MSE: 0.201262, Test Acc: 0.7187
Epoch 6200/10000 - Train MSE: 0.203558, Test MSE: 0.201253, Test Acc: 0.7187
Epoch 6300/10000 - Train MSE: 0.203552, Test MSE: 0.201246, Test Acc: 0.7187
Epoch 6400/10000 - Train MSE: 0.203546, Test MSE: 0.201240, Test Acc: 0.7187
Epoch 6500/10000 - Train MSE: 0.203541, Test MSE: 0.201234, Test Acc: 0.7186
Epoch 6600/10000 - Train MSE: 0.203536, Test MSE: 0.201229, Test Acc: 0.7186
Epoch 6700/10000 - Train MSE: 0.203531, Test MSE: 0.201225, Test Acc: 0.7186
Epoch 6800/10000 - Train MSE: 0.203527, Test MSE: 0.201221, Test Acc: 0.7186
Epoch 6900/10000 - Train MSE: 0.203523, Test MSE: 0.201218, Test Acc: 0.7187
Epoch 7000/10000 - Train MSE: 0.203520, Test MSE: 0.201215, Test Acc: 0.7187
Epoch 7100/10000 - Train MSE: 0.203516, Test MSE: 0.201213, Test Acc: 0.7187
Epoch 7200/10000 - Train MSE: 0.203513, Test MSE: 0.201211, Test Acc: 0.7187
Epoch 7300/10000 - Train MSE: 0.203510, Test MSE: 0.201209, Test Acc: 0.7187
Epoch 7400/10000 - Train MSE: 0.203507, Test MSE: 0.201207, Test Acc: 0.7187
Epoch 7500/10000 - Train MSE: 0.203505, Test MSE: 0.201206, Test Acc: 0.7187
Epoch 7600/10000 - Train MSE: 0.203502, Test MSE: 0.201205, Test Acc: 0.7184
Epoch 7700/10000 - Train MSE: 0.203500, Test MSE: 0.201204, Test Acc: 0.7184
Epoch 7800/10000 - Train MSE: 0.203498, Test MSE: 0.201203, Test Acc: 0.7184
Epoch 7900/10000 - Train MSE: 0.203496, Test MSE: 0.201202, Test Acc: 0.7184
Epoch 8000/10000 - Train MSE: 0.203494, Test MSE: 0.201201, Test Acc: 0.7184
Epoch 8100/10000 - Train MSE: 0.203492, Test MSE: 0.201201, Test Acc: 0.7184
Epoch 8200/10000 - Train MSE: 0.203490, Test MSE: 0.201200, Test Acc: 0.7184
Epoch 8300/10000 - Train MSE: 0.203488, Test MSE: 0.201200, Test Acc: 0.7184
Epoch 8400/10000 - Train MSE: 0.203486, Test MSE: 0.201200, Test Acc: 0.7184
Epoch 8500/10000 - Train MSE: 0.203485, Test MSE: 0.201199, Test Acc: 0.7184
Epoch 8600/10000 - Train MSE: 0.203483, Test MSE: 0.201199, Test Acc: 0.7184
Epoch 8700/10000 - Train MSE: 0.203482, Test MSE: 0.201199, Test Acc: 0.7184
Epoch 8800/10000 - Train MSE: 0.203480, Test MSE: 0.201198, Test Acc: 0.7184
Epoch 8900/10000 - Train MSE: 0.203479, Test MSE: 0.201198, Test Acc: 0.7184
Epoch 9000/10000 - Train MSE: 0.203477, Test MSE: 0.201198, Test Acc: 0.7184
Epoch 9100/10000 - Train MSE: 0.203476, Test MSE: 0.201198, Test Acc: 0.7184
Epoch 9200/10000 - Train MSE: 0.203475, Test MSE: 0.201198, Test Acc: 0.7184
Epoch 9300/10000 - Train MSE: 0.203473, Test MSE: 0.201198, Test Acc: 0.7184
Epoch 9400/10000 - Train MSE: 0.203472, Test MSE: 0.201198, Test Acc: 0.7184
Epoch 9500/10000 - Train MSE: 0.203471, Test MSE: 0.201197, Test Acc: 0.7184
Epoch 9600/10000 - Train MSE: 0.203470, Test MSE: 0.201197, Test Acc: 0.7184
Epoch 9700/10000 - Train MSE: 0.203468, Test MSE: 0.201197, Test Acc: 0.7184
Epoch 9800/10000 - Train MSE: 0.203467, Test MSE: 0.201197, Test Acc: 0.7184
Epoch 9900/10000 - Train MSE: 0.203466, Test MSE: 0.201197, Test Acc: 0.7184
Epoch 10000/10000 - Train MSE: 0.203465, Test MSE: 0.201197, Test Acc: 0.7184

Best epoch: 10000

Best test MSE: 0.201197